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Early-age behaviour of the concrete surrounding a turbine spiral case: Monitoring and thermo-mechanical modelling

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ABSTRACT

Within massive structures hydration reactions of the cement present in the mixture promote an internal heat release. Due to the low thermal conductivity of concrete, the dissipation rate of this thermal energy is lower than the rate of production of hydration heat, so considerable temperature gradients occur. Therefore, thermal stresses arise as a consequence of differential thermal expansions or contractions. These self-induced stresses may be even higher in the scenario of additional external restraints (such as support conditions). If not adequately controlled, the self-induced tensile stresses may induce cracking of concrete at early ages. This paper describes the in-depth analysis of the concrete surrounding the spiral case of a hydroelectric turbine constructed in Brazil, with a comprehensive approach that includes: laboratory characterization of the concrete properties, numerical analyses to predict the temperature and stress fields, and the corresponding in-field monitoring. The consistency obtained between numerical results and *in situ* observations is discussed in detail. The thermally induced cracking risk is also evaluated.

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1. Introduction

The exothermic chemical reactions that occur in concrete due to cement hydration at early ages foment high temperature gradients between the core regions and the surfaces of the structural elements. Due to the induced differential concrete strains on the core and surface regions, as well as to external restraints, these temperature gradients may lead to tensile stresses that reach the tensile strength of the young concrete, and thereby cause thermal cracking which jeopardizes concrete durability, has detrimental effects on aesthetics, and may even limit the intended service-life performance (e.g. leakage in a reservoir). These effects are prominent in massive concrete structures, such as bridge footings, nuclear containments and dams. Evaluation of the thermo-mechanical behaviour of massive concrete during construction is a relevant challenge that structural engineers may be faced with, as demonstrated by the several attempts dealing with this issue published in recent years [1–8].

In hydroelectric power plants the powerhouse substructure supports the turbines and generators, as well as the superstructure. The substructure contains the water passages, includes rooms and galleries needed for mechanical and electrical equipment, and provides the mass needed for stability (see Fig. 1a) [9]. Normally, in the vicinity of the steel spiral cases, it is necessary to guarantee recesses in the substructure for accommodation of the spirals, and to devise the construction phasing of the power plant in such a way that concreting operations can continue without interruption during installation of the spiral components [9]. Therefore, the overall concrete placement is made in two stages: first, all the elements that do not contain turbine parts are cast, and afterwards the concrete surrounding the steel spiral is placed. Generally, the substructure concrete is placed in lifts 1.5 m thick or less, in order to minimize thermal cracking; on the other hand, massive concrete placements are necessary where concrete needs to be watertight, such as in spiral cases [9]. The significant volumes of concrete involved in the second phase necessarily lead to large amount of heat release due to the exothermic nature of cement hydration reactions. Temperature gradients between the core and the surface regions of the hardening concrete are thus expected. In addition, the hardened concrete of the first stage restrains the deformation of the hardening concrete of the second stage.

Numerical predictions of the temperature and stress fields in the concrete surrounding spiral cases were published by few

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authors in recent years [10–13], but so far comparisons of numerical predictions against monitored results are still lacking. Moreover, an accompanying extensive characterization of the concrete properties by laboratory tests, such as thermal diffusivity, rate of cement hydration, elastic E-modulus growth, tensile strength, basic creep and autogenous deformation, has not yet been presented in this context. The present paper aims to provide further information about the thermo-mechanical performance of this type of massive structures, based on the real study case of the Batalha Hydroelectric Power Plant (Batalha HPP), built in the San Marcos River (Brazil), in which the concrete around the spiral case was monitored during the construction phase. To this end, the following research methodology was applied: (i) characterization of concrete properties via laboratory testing, (ii) *in situ* monitoring of concrete temperatures and strains, (iii) thermo-mechanical modelling of the concrete that encases the steel spiral and evaluation of the concrete cracking risk. The concrete spiral case under investigation is made up of lightly reinforced concrete (massive concrete), with dimensions of 13.90 m × 14.15 m × 5.83 m (see Fig. 1b). The steel spiral case was embedded in concrete under pressurized conditions, i.e., the spiral was filled with water and pressurized to the test pressure, before the pouring of concrete. The thermo-mechanical modelling described in this paper consists on the initial computation of the thermal field, and subsequent evaluation of the corresponding strain field. The thermal problem was reproduced using a transient model, which allows determining concrete temperatures taking into account the internal heat generated by cement hydration reactions, the thermal conductivity and specific heat of concrete, the heat losses towards the surrounding environment, and the influence of the temperature of the water encased inside the spiral.

The time-dependent mechanical behaviour of concrete is described by the theory of linear viscoelasticity for aging materials. The mechanical problem encompasses the evolution of concrete mechanical properties, as well as the influence of thermal and autogenous deformations and basic creep. This thermo-mechanical methodology, based on the Finite Element Method (FEM), is similar to the ones reported in [14–16]. Special attention is paid to the material properties and modelling assumptions, which are validated by comparing the predicted temperatures and strains with the monitored results. Finally, evolutions of the concrete stresses and corresponding cracking risks are analysed in detail.

2. Thermal–mechanical modelling

2.1. Thermal problem

The evolution of temperatures is calculated from the heat balance equation [17]

$$k\nabla\cdot(\nabla T)+\dot{Q}=\rho c\dot{T} \tag{1}$$

where k is the thermal conductivity, T is the temperature, $\dot{Q}$ is the rate of internal heat generated by cement hydration and ρc is the volumetric specific heat. At a macro-scale, that is, at the engineering level at which large real concrete structures are analysed, the rate of internal heat generation, which reflects the thermo-activated nature of cement hydration reactions, can be expressed with an Arrhenius-type law [18]:

$$\dot{Q}=a\,f(\alpha)\,e^{-E_a/(RT)} \tag{2}$$

where a is a constant, $f(\alpha)$ is a function representing the normalized evolution of the heat production rate, the hydration degree α is the ratio between the total heat $Q(t)$ generated until instant t and the total heat that can be released upon completion of hydration reactions Q_∞, E_a is the apparent activation energy and R is the universal gas constant (8.314 J mol^{-1} K^{-1}). In the application of this paper E_a is taken as a constant, being denoted as 'apparent' as it provides a simplified macro representation of the complex set of individual chemical reactions that really occur at a micro-scale level in cement hydration (a more complete representation of those individual chemical reactions is performed by several authors, like in Refs. [19–21]). The simplification of considering E_a as a constant is backed by the proposal of Wadsö [22] and validated with experimental results by Azenha [23].

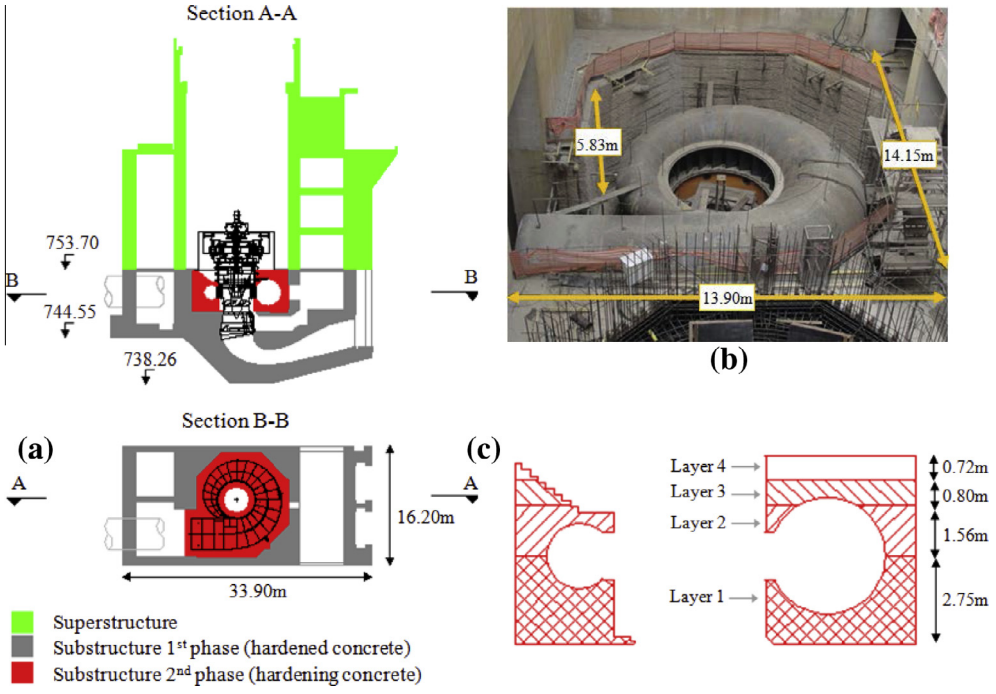


Fig. 1. Batalha hydroelectric power plant: (a) overall structure; (b) spiral case; (c) casting layers of the spiral case (section A-A).

Boundary conditions are described by the Newton's cooling law [24]

$$q = h(T_b - T_a) \tag{3}$$

where q is the heat flow per unit area, h is the convection coefficient, T_b is the temperature at the surface boundary and T_a is the air temperature. Usually, convection and radiation effects are reproduced together with a single convection-radiation coefficient h_{cr}, which is used in Eq. (3) instead of h [14].

In practical applications there may exist relatively thin layers of materials between the air and the concrete surfaces, in correspondence to insulation or formworks. In these cases the coefficient of convection-radiation may be calculated by an equivalent coefficient of heat transfer h_{eq} [14]

$$h_{eq} = \left(\frac{1}{h_{cr}} + \sum_{i=1}^{n} \frac{L_i}{k_i}\right)^{-1} \tag{4}$$

where L_i represents the thickness and k_i the thermal conductivity of the material of layer i.

Through implementation of the heat balance Eq. (1) in a FEM code, and adopting a backward-Euler time-stepping integration scheme, computation of the transient temperature field is carried out. A remark is made to the fact that Eq. (1) leads to a nonlinear solving problem, due to the dependency of the heat rate $\dot{Q}$ on the temperature T, expressed in Eq. (2). Accordingly, an incremental-iterative procedure supported by the Newton–Raphson method is adopted to solve the global FEM system of equations. To find details concerning the FEM implementation, the incremental-iterative Newton-Raphson procedure and the individual calculation steps the reader is addressed to Ref. [14].

2.2. Mechanical problem

The mechanical model is activated after the thermal one, from which it receives the concrete temperatures necessary for computing the local strains and stresses. Therefore, the same time advancing scheme adopted for the thermal analysis is usually implemented. The stress calculations are made based on the theory of linear viscoelasticity for aging materials, and evolution of the concrete mechanical properties as consequence of cement hydration is quantified using the Maturity Method [25], based on the equivalent age t_{eq} concept,

$$t_{eq} = \int_0^t e^{-\frac{E_a}{R}\left(\frac{1}{T(\tau)} - \frac{1}{T_{ref}}\right)} d\tau \tag{5}$$

where T_{ref} is the reference temperature of the laboratory tests and t is the actual time.

The pronounced concrete basic creep that occurs during early ages is reproduced by the Double Power Law [26]

$$J(t,t') = \frac{1}{E_0(t')} + \frac{\phi_1}{E_0(t')}(t')^{-m}(t-t')^n \tag{6}$$

where $J(t,t')$ is the compliance function at time t for a load applied at instant t', $E_0(t')$ is the asymptotic elastic modulus (corresponding to short term loads), and ϕ_1, m and n are material parameters.

Assessment of the local cracking risk of concrete, $CR(t)$, is evaluated by computing the ratio between the maximum principal concrete tensile stress calculated at an integration point at instant t, $\sigma_1(t)$, and the estimated concrete tensile strength $f_{ct}(t)$ at the same location and time.

The described thermal–mechanical model is implemented in the FEM computational code DIANA [27], which was used for the analyses presented in this paper. A detailed flowchart of the calculation scheme, including the individual calculation steps, is provided in Ref. [14].

3. Application: concrete spiral case of Batalha HPP

3.1. Overview

The thermo-mechanical framework described in Section 2 will be used to predict temperatures, strains and stresses on the concrete placed around the spiral case of one of the Kaplan turbines installed on the powerhouse of Batalha HPP, built between the Brazilian states of Goiás and Minas Gerais. The overall dimensions of the concrete block are indicated in Fig. 1b. The concreting operation was divided in four layers (see Fig. 1c), with thicknesses of 2.75 m, 1.56 m, 0.80 m and 0.72 m, respectively (from bottom to top). The first layer had a concrete volume of 251 m^3, which was poured during 11 h, approximately. Each of the remaining layers mobilized a concrete volume of about 100 m^3. The execution of the upper layers was delayed by on-site constraints, and no precise information about their concreting dates was provided by the construction management team. In this paper the analysis is focused on the first layer, since it is the thickest and potentially shall present a higher cracking risk. It is also important to emphasize that the study of the first layer is restricted to a period of 17 days, strictly for which monitoring data is available. The influence of the casting operations of the upper layers is not considered in this analysis, as they had not yet been cast when the monitoring was stopped (at the age of 17 days). The 3D layout of the spiral casing is depicted in Fig. 2, where it is also possible to see the water contained in the steel spiral, the hardened concrete within which the hardening concrete was poured, as well as part of the supporting soil.

3.2. Monitoring campaign

During construction a monitoring campaign was carried out, using 6 Carlson strain gages that were installed in the first casting layer, to assess concrete temperatures and strains in specified points. Five of them were installed in section 1-1 (see Fig. 3). All the sensors were positioned perpendicularly to section 1-1, with the exception of sensor S4 that was placed in the plane of section 1-1, at an angle of 45° with the horizontal. Sensor S1 was placed 0.20 m above the bottom hardened concrete, and vertically aligned with sensor S2, which was installed at the location where the concrete temperature was expected to reach a maximum.

In order to evaluate autogenous and thermal deformations of the hardening concrete, strains were also measured on a 'no-stress box'. The 'no-stress box' consists of a small mould, filled with fresh concrete, and embedded within the structural element that is being cast. The mould is devised in such a way that it allows thermal continuity between the inner and the surrounding concrete, whilst it assures full mechanical detachment between the inner and the outer concrete – see Fig. 4. More details about the working principles and assembly of a 'no-stress box' can be found in Refs. [28,29]. Hence, sensor S3 was placed inside the 'no-stress box', with its longitudinal axis placed 0.10 m above sensor S2.

In order to monitor the influence of the proximity to the spiral, sensor S4 was placed inside the concrete, 0.20 m away from the steel surface of the spiral. Sensor S5 was installed 0.23 m away from the vertical hardened concrete wall, towards the interior of the concrete layer, in order to assess the restraining influence of that boundary. Due to construction-related reasons, sensor S6 (see Fig. 3a) had to be installed outside section 1-1, halfway between the hardened concrete and the steel spiral, and 0.20 m below the top surface in order to capture the influence of the

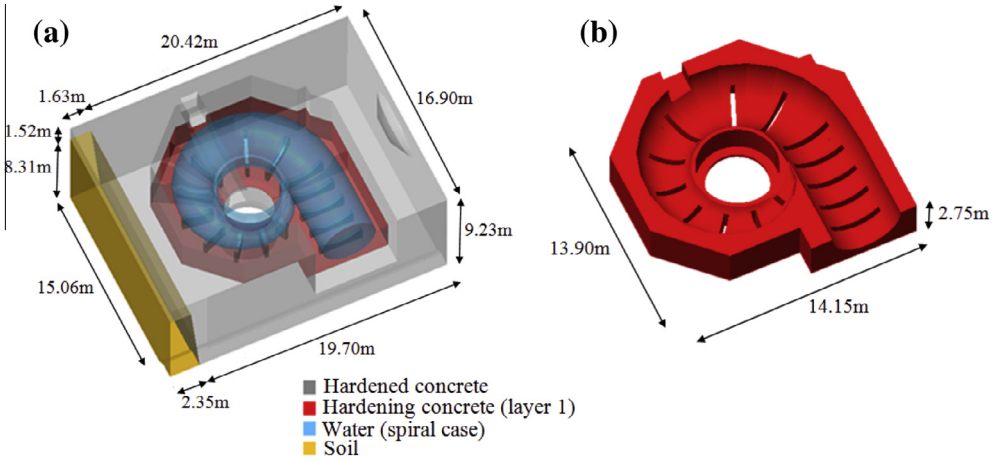


Fig. 2. 3D layout of the spiral casing: (a) general view; (b) layer 1.

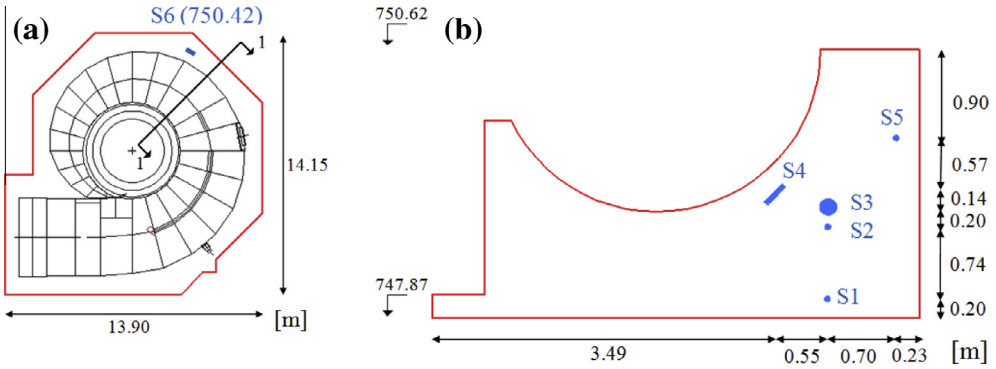


Fig. 3. Monitored section: (a) plan view (EL 750.62); (b) Carlson sensors in section 1-1.

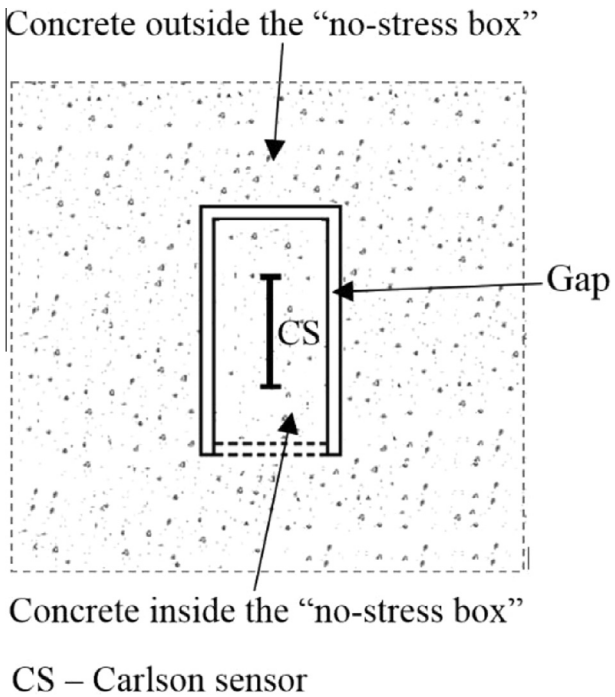


Fig. 4. Schematic representation of the ‘no-stress box’ principle.

environmental temperature. The sensors measured temperatures and strains since the beginning of concrete placement, and measurements were taken over a period of 17 days.

The environmental temperature was measured in the vicinity of the spiral case, but it could not be continuously recorded, and the 43 discrete measurements at the construction site were made with a considerable irregularity in terms of the instant at which they were taken. This induced some spurious scatter, because the different measurements were not related to the same fixed hour on every day. Fortunately, as the amplitudes of the environmental temperatures were small (less than 6.6 °C), they will endure ambient temperature variations with little influence for the thermal analyses, regarding the small thermal conductivity of concrete. Accordingly, an average daily temperature of 25 °C was assumed instead. It should be noted that consideration of an average daily temperature at the site of a structure is also recommended by the latest guidelines from JCI – Japanese Concrete Institute [30] for thermo-mechanical modelling of mass concrete.

3.3. Modelling strategy and FE meshes

The temperature and stress fields in the concrete surrounding the spiral were modelled with a staggered heat flow-stress analysis. A 2D idealization (coincident with section 1-1) and a refined 3D modelling were performed. Fig. 5 reproduces the geometry adopted for the 2D model, which includes the layer of hardening concrete, as well as the hardened concrete and the water contained inside the spiral. The 3D model, whose boundary conditions are reproduced in Fig. 6 includes also part of the supporting soil and the draft tube, as well as the access galleries, which were neglected in the 2D model.

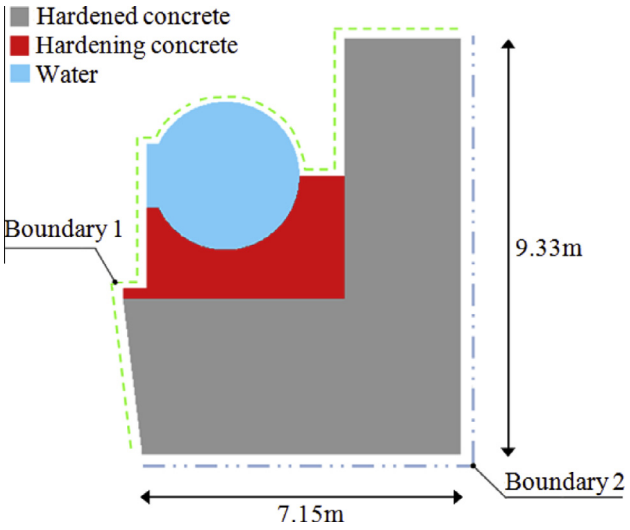


Fig. 5. Schematic representation of the 2D model (coincides with section 1-1).

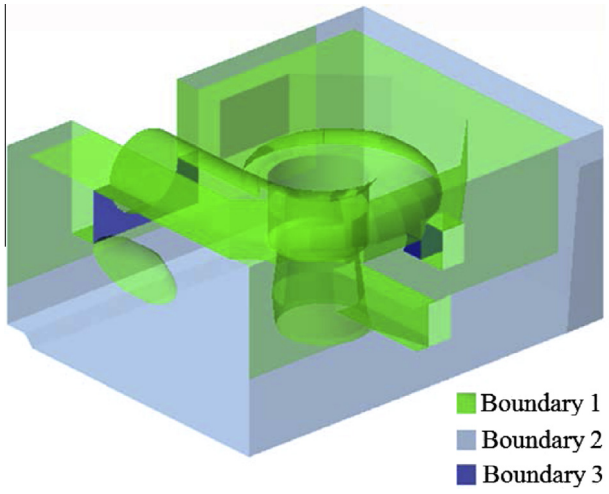


Fig. 6. Thermal boundaries of the 3D model.

Explicit modelling of the embedded reinforcement was disregarded, for two main reasons: (i) the thermal compatibility with concrete due to the similarity of the coefficients of thermal expansion of both materials [30]; (ii) the low restraint that rebars represent to concrete in massive structures.

The steel spiral was also not explicitly incorporated in the analyses. From a thermal point of view, this modelling assumption is valid due to the high conductivity of steel and the reduced thickness of the spiral case. As far as the mechanical model, this simplification is also acceptable because only the first concrete layer is to be modelled – see Fig. 5 –, and accordingly only just the bottom-half of the spiral is in contact with the concrete. Therefore, during this construction stage one does not have the steel spiral fully embedded in concrete, and so the former does not significantly restrict the concrete thermal expansions and contractions, as also confirmed on parametric analyses performed for this purpose on the 2D model (not presented here for the sake of conciseness), with and without explicit consideration of the steel spiral. In addition, it should be noted that the non-inclusion of the steel spiral is in line with the modelling strategy adopted for similar structures in previous works [10,13].

Concerning the thermal models three boundary conditions were considered, with a nomenclature that is applicable to both 2D and 3D models (see Figs. 5 and 6 and Table 1):

- *Boundary 1:* all materials (steel spiral, hardening concrete and hardened concrete) are assumed to be in direct contact with air. The corresponding convection-radiation coefficient was considered equal to $h_{eq} = 10\ Wm^{-1}\ K^{-1}$, reproducing an almost stagnant air condition, in view of studies reported in [23,31].
- *Boundary 2:* reproduces the distant hardened concrete and the supporting soil, across which heat fluxes were assumed to be null (adiabatic condition);
- *Boundary 3 (only in 3D model):* surfaces corresponding to 20 mm thick wood formworks, where $h_{eq} = 4.3\ Wm^{-1}\ K^{-1}$ was estimated from Eq. (4).

Around boundaries 1 and 3 a constant average daily temperature of 25 °C was taken as input to the numerical model, as referred before. During casting operations the initial temperature for the hardening concrete was taken as 32 °C, in accordance to the Carlson sensor measurements right after being in contact with the hardening concrete. For the remaining materials (hardened concrete, water and soil) an initial temperature of 25 °C was used, thus assuming thermal equilibrium with the environment. The influence of solar radiation was disregarded, given that the spiral case is located inside the power plant, and remained shadowed from direct sunlight.

With regards to the mechanical boundary conditions, all degrees of freedom (d.o.f.) were restrained on boundary 2, as indicated in Table 1. Along the other boundaries no restrictions were considered to the degrees of freedom.

In the 2D model the transient heat flow, analyses were performed using 3-noded isoparametric FE for the concrete and 2-noded FE for the boundaries; for the structural analyses 6-noded (axisymmetric) and 3-noded FE were adopted for the concrete and boundaries, respectively. In the 3D model the transient heat flow analyses were performed with 4-noded isoparametric FE tetrahedrons for the concrete and 3-noded FE triangles to describe the boundaries; the mechanical analyses were performed using 10-noded isoparametric FE tetrahedrons for the concrete and 6-noded FE for the boundaries. The adopted FE meshes are reproduced in Figs. 7 and 8.

The performed thermo-mechanical analyses spread over a total period of 17 days – the strict period along which monitoring data was available – using 28 time steps ranging from 0.2 days at the beginning of the analyses up to 1 day at the end. These time steps were defined based on several tentative analyses, aiming to keep the computational times within feasible limits (in the 3D analyses a maximum of 12 h was consumed), but without losing the required accuracy, especially during the first days after casting.

3.4. Material properties

The concrete mixture used in the actual application has the composition summarized in Table 2. In order to characterize the heat development of the cement paste, isothermal calorimetric tests at several temperatures (15 °C, 25 °C, 35 °C, 45 °C and 55 °C) were performed using a TAM Air calorimeter. The total heat Q_∞ was determined from the curves shown in Fig. 9a, in accordance with the method presented in [23]. The evolution of Q_∞ for different isothermal temperatures is given in the following list, in which the information is provided within brackets as [T(°C); Q_∞ (kJ/kg)]: [15; 269.25], [25; 274.61], [35; 288.15], [45; 293.18] and [55; 317.79]. The test results show a rise of Q_∞ by increasing the testing temperature, a tendency also reported in [30,33] for cements blended with blast-furnace slag. The results obtained for $f(\alpha)$ in the experimental campaign are depicted in Fig. 9b, where the existence of two peaks is clear: the first one corresponds to the Portland cement hydration, while the second concerns to the slag reaction. This behaviour was also reported in [34]. The secondary slag

Table 1
Thermal and mechanical boundary conditions.

Model	Boundary	Thermal	Mechanical BC
2D	1	Convection	–
	2	Adiabatic	All d.o.f. fixed
3D	1	Convection	–
	2	Adiabatic	All d.o.f. fixed
	3	Convection	–

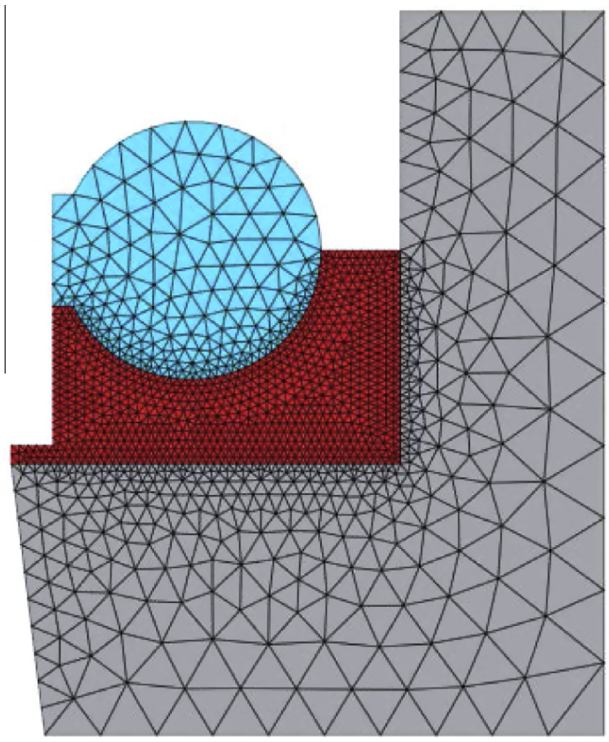


Fig. 7. Axisymmetric FE mesh.

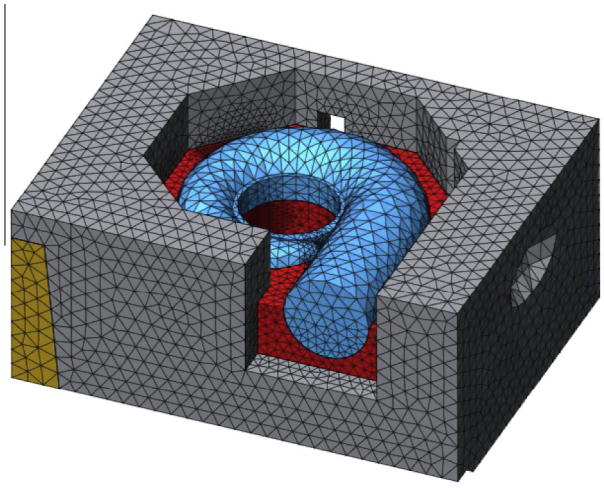


Fig. 8. 3D FE mesh.

reaction is more sensitive to temperature than the Portland cement one, but there is a stabilization of $f(\alpha)$ for 45 °C and 55 °C. The apparent activation energy was calculated from the so-called 'Derivative Speed Method' [23], resulting in E_a/R = 4391.3 K. Based on these experimental results, the parameters for the heat generation rate $\dot{Q}$ for the studied concrete are shown in Fig. 10.

Table 2
Concrete mix composition.

Cement CP III 40 RS [32] (60% is blast-furnace slag)	273 kg/m^3
Water	196 kg/m^3
Natural sand	815 kg/m^3
Gravel 25 mm (limestone)	1025 kg/m^3
Water-to-binder ratio (w/b)	0.72

By making use of the reference values proposed by Breugel [35], thermal conductivity and specific heat of the utilized concrete were estimated based on a ponder average of the thermal properties of the mixture components, resulting in the following values: k = 3.0 Wm^{-1} K^{-1} and ρc = 2420 kJ m^{-3} K^{-1}. These values result in a thermal diffusivity of 0.107 m^2/day, which is consistent with the experimental value (0.106 m^2/day) assessed according to [36].

In line with the isothermal calorimetric tests mentioned above, an adiabatic calorimeter test was conducted (for an initial temperature of 23 °C, the Brazilian reference temperature prescribed for concrete experimental characterization), which resulted in an ultimate adiabatic temperature rise of 26.4 °C (see Fig. 11). Using ρc = 2420 kJ m^{-3} K^{-1}, the curves obtained in the isothermal tests (Fig. 9a) can be transformed into corresponding adiabatic ones, by using Eq. (1) for determining the temperature field within the adiabatic chamber and considering the volumetric content of cementitious matrix in the mixture. As shown in Fig. 11, the adiabatic curve at 23 °C and the adiabatic curve calculated from the isothermal calorimetry test at 25 °C agree very well.

For the mechanical characterization of concrete, several specimens were cast to evaluate the E_c-modulus (Fig. 12) and the tensile strength (Fig. 13) at several ages under a curing temperature of 23 °C according to [37,38], respectively. Compressive creep tests were also carried out (basic creep) using cylinders with a diameter of 15 cm and a height of 50 cm, and loaded at different ages (3, 7 and 28 days), following [39]. The specimens for creep testing were sealed to avoid moisture losses and drying shrinkage under a curing temperature of 23 °C, and the applied load induced a compressive stress of ~40% of the concrete compressive strength at each age of loading (in order to assure a linear viscoelastic regimen). The Double Power Law (DPL) was then fitted to the experimental results of creep testing, as illustrated in Fig. 14.

The autogenous volumetric deformations of concrete were monitored (through Carlson strain meters) in three sealed specimens with dimensions 10.0 cm × 10.0 cm × 29.4 cm, under a curing temperature of 30 °C (see Fig. 15). Using the equivalent age concept, and applying the apparent activation energy calculated by calorimetry testing, the autogenous deformation was thereafter numerically evaluated for a reference temperature of 23 °C. In fact, the observed autogenous deformations during the first 20 days correspond to a volumetric expansion (swelling) – a peculiar behaviour that was also reported in [40,41] for cement pastes with high water/cement ratios, like in the present concrete – followed by a contraction (shrinkage) after that period.

The coefficient of thermal expansion of concrete α_{cT} has been reported to endure significant time-dependent variation at very early ages [42–44]: it exhibits high values upon setting (20×10^{-6} °C^{-1} or even higher), decreasing rapidly to about 10×10^{-6} °C^{-1} at 12–14 h after setting. Nevertheless, the deviations introduced in numerical results by adopting a constant value of α_{cT} are considered to be acceptably low for the calculation of stresses, since the E_c-modulus of concrete is quite low during the first hours, and thus are the corresponding induced stresses in concrete. The total deformation measured by sensor S3 located at the 'no-stress box' (Fig. 16a) corresponds to the sum of the free thermal and autogenous deformations of concrete (TD and AD,

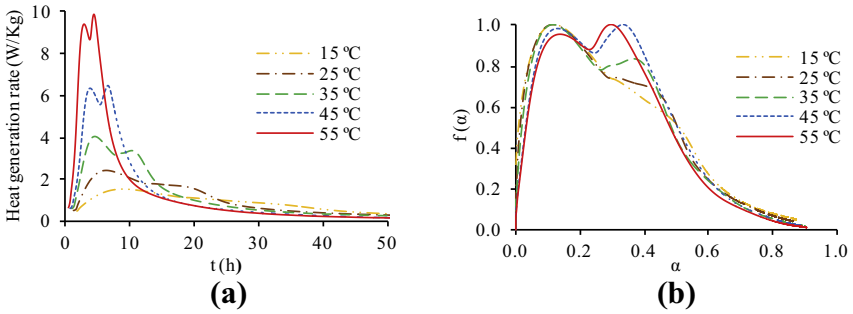


Fig. 9. Isothermal calorimeter testing: (a) heat generation rate per kg of cement; (b) normalized heat.

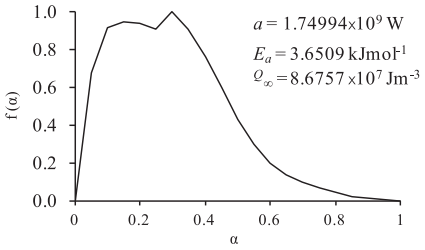


Fig. 10. Arrhenius law for the concrete – $\dot{Q}$.

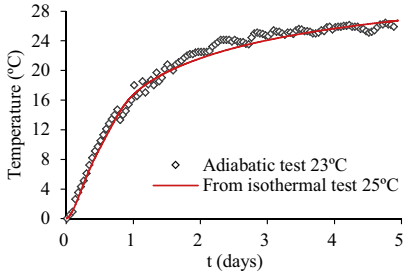


Fig. 11. Adiabatic temperature rise.

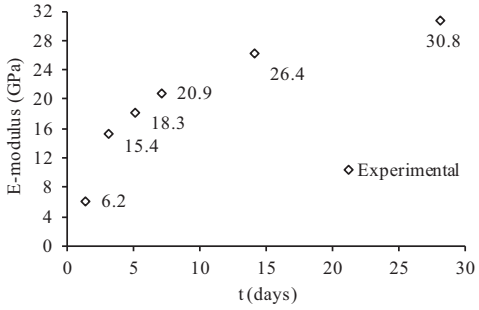


Fig. 12. Evolution of concrete E_c-modulus.

respectively). In Fig. 16b the TD deformation was obtained by subtracting the AD average deformation measured on the concrete sealed specimens and reproduced in Fig. 15 (duly corrected for maturity effects) from the TD + AD deformation reproduced in Fig. 16a. Based on this procedure, from Fig. 16b an average α_{cT} of about 10×10^{-6} °C^{-1} was determined for the entire heating phase and along the subsequent cooling phase. The constant value of $\alpha_{cT} = 10 \times 10^{-6}$ °C^{-1} was thus used in all the numerical analyses. Poisson's coefficient was adopted as $\upsilon = 0.2$.

The material properties of the hardened concrete were considered the same as those assumed for hardening concrete, but with

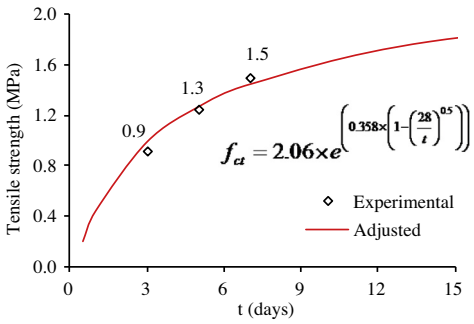


Fig. 13. Evolution of concrete tensile strength.

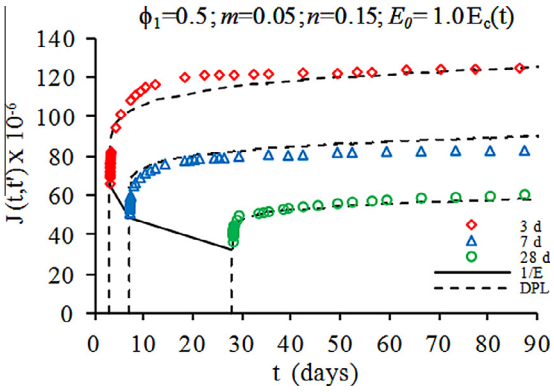


Fig. 14. Concrete basic creep curves and DPL coefficients.

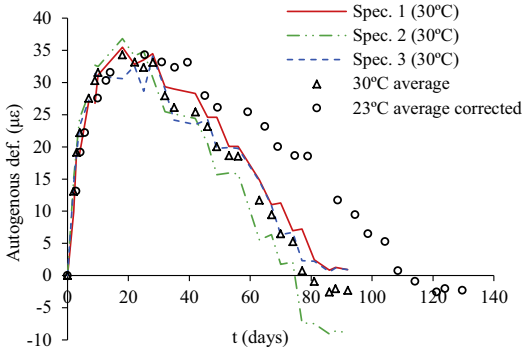


Fig. 15. Autogenous deformation: measured strains (lines) and average strain value at 30 °C, as well as at the corrected temperature of 23 °C.

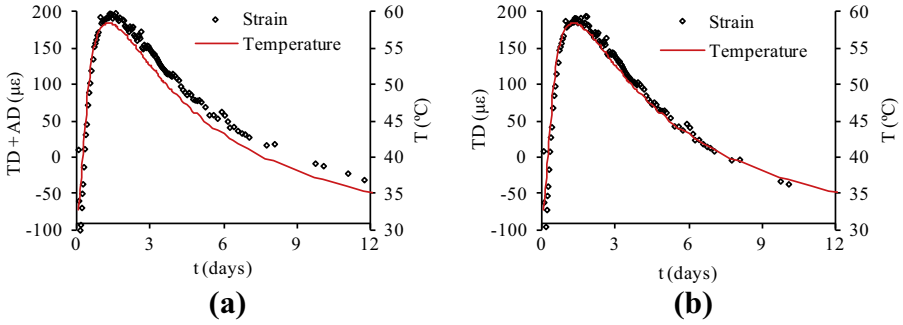


Fig. 16. Monitored results at sensor S3 (no-stress box): (a) TD+AD; (b) TD.

no regard to the cement hydration. Besides, the viscoelastic behaviour corresponding to an age of loading of 28 days was adopted for the hardened concrete.

For the supporting soil, constituted by quartz phyllites, the following thermal properties were assumed: $k = 2.87\ \mathrm{Wm^{-1}\,K^{-1}}$ and $\rho c = 2261\ \mathrm{kJ\,m^{-3}\,K^{-1}}$ [45]. According to the geotechnical survey [46], the *E*-modulus of the soil was ~1.1GPa, which is coherent with the results reported in [47] for the same type of soil; besides, a Poisson's coefficient of $\upsilon = 0.2$ and a coefficient of thermal expansion of $\alpha_{cT} = 10 \times 10^{-6}\ ^{\circ}\mathrm{C}^{-1}$ were adopted for the soil.

In the case of the water contained in the spiral, its internal heat flux is governed by conduction and natural convection processes. According to [17], the conduction/convection heat transfer in an enclosed region can be evaluated through a simple conduction model in which the thermal conductivity k is replaced by $k \times N_u$ (N_u is the Nusselt number), as a result of convection currents. The $k \times N_u$ is called the 'effective thermal conductivity of the enclosure'. This is a very convenient simplification, as the natural convection heat transfer depends on many factors, such as: geometry of the enclosure, its orientation, variation of temperature on the surfaces, and the properties of the fluid involved. In the present study, the geometric complexity of the enclosure makes it difficult to obtain reliable estimates for N_u based on simple analytical relations available in the literature, since they are assessed for very simple geometries. Therefore, the effective thermal conductivity was determined by a trial-and-error process, so that the temperatures in the numerical model during the cooling phase coincided with the temperatures registered by sensor S4, which is the closest to the water. Thus, the thermal conductivity of water was set to $k \times N_u = 4.0\ \mathrm{Wm^{-1}\,K^{-1}}$ (a significantly different value from the standard conductivity of $0.6\ \mathrm{Wm^{-1}\,K^{-1}}$ for stagnant water).

The remaining properties of water were taken as: $\rho c = 4187\ \mathrm{kJ\,m^{-3}\,K^{-1}}$, $\alpha_{Tw} = 100 \times 10^{-6}\ ^{\circ}\mathrm{C}^{-1}$, volumetric deformability modulus $K_w = 2.11$ GPa and shear elasticity modulus $G_w = 0$ MPa.

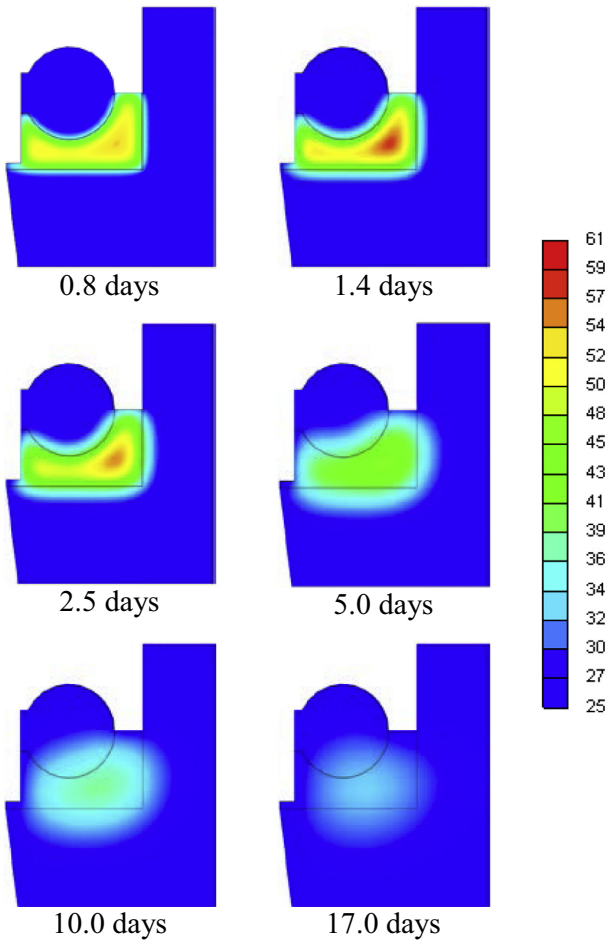


Fig. 17. 2D numerical prediction of temperatures (°C) in section 1-1.

3.5. Numerical results and comparison with the monitored temperatures/strains

3.5.1. Temperatures

The 2D numerical predictions of concrete temperatures in section 1-1, at different ages, are reproduced in Fig. 17 (see also Fig. 18 for completeness). The heating period starts immediately after concrete placement, and lasts about 1.4 days, instant at which the maximum temperature of ~58 °C is reached on the centre of the concrete layer. Then the high temperatures in the core region of the hardening concrete dissipate through the external boundaries and neighbouring bodies (water and hardened concrete). It can be noticed that even after 17 days, thermal equilibrium with the surrounding environment has not yet been reached, due to the considerable volume of the layer and the low conductivity of concrete.

Comparisons between the numerically predicted and the monitored temperatures are shown in Fig. 18, where the results from the 2D and 3D models are plotted as lines, whereas experimental results obtained at the several Carlson sensors are marked by small rhombus. In general, very good agreement between the numerical and the monitored temperatures can be observed, thus validating the modelling assumptions that were considered. It can also be observed that the differences between the temperatures predicted by the 2D and 3D numerical models are very small, always under 1 °C. The discrepancies in the temperature peaks predicted for sensors S1 and S5 (see Fig. 18a and d) may be explained by the

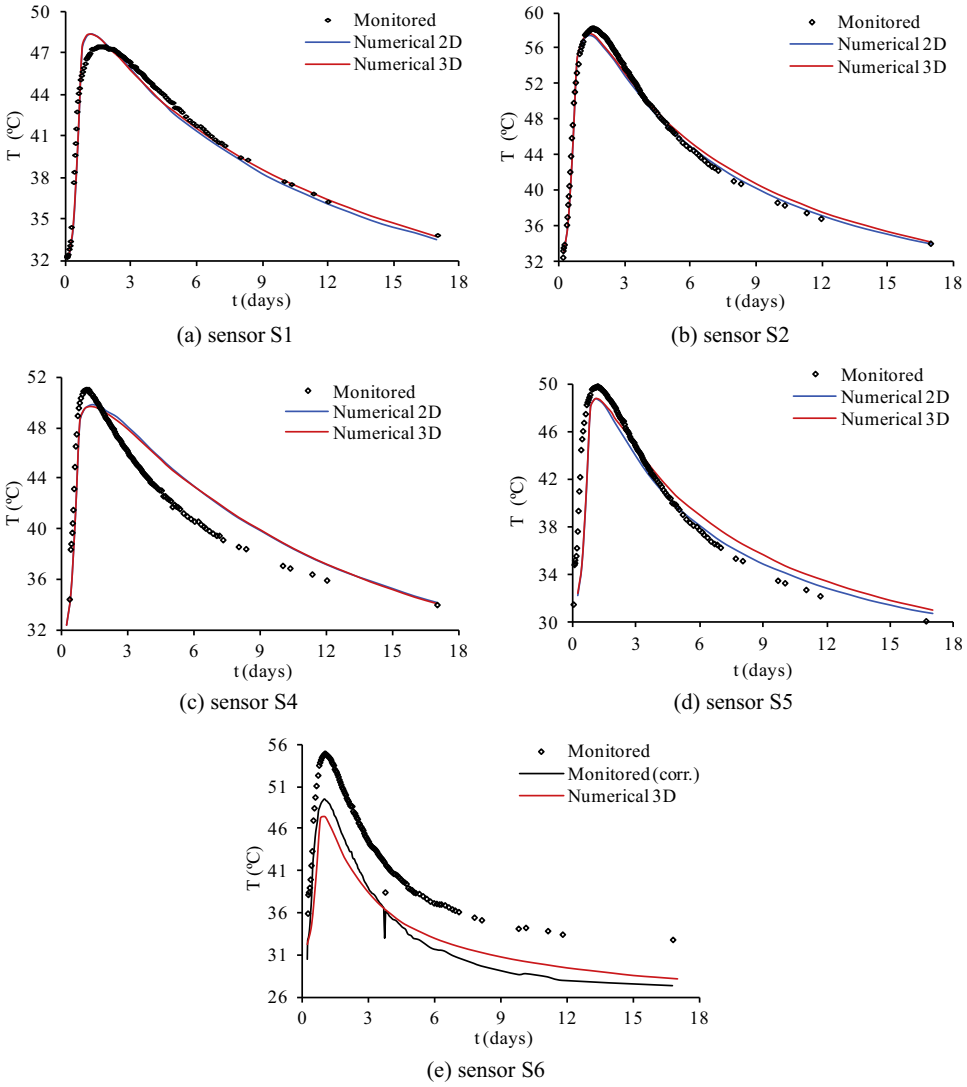


Fig. 18. Temperatures: numerical vs. monitored.

uniform temperature field simplification, assumed as the initial condition for the hardened concrete. For sensor S2 (Fig. 18b), a very good coherence is observed, as the monitored peak temperature of ~58 °C was correctly predicted by the numerical model. In the case of sensor S4 (Fig. 18c) a reasonable coherence is noticed, as there is an error of ~1.5 °C in the peak temperature value; however, the cooling period was not so correctly predicted, which is due to the fact that a constant value was assumed for the Nusselt number (and thus for the thermal conductivity of water), when in reality N_u varies in time, as it depends on the temperatures at the surfaces of the spiral case. Nevertheless, the numerical results obtained for the other sensors certify the reasonableness of the adopted modelling strategy.

With regards to Carlson sensor S6, an unexpected trend was detected (see Fig. 18e): this sensor exhibits higher initial temperatures when compared to the other sensors. This discrepancy could be explained by a possible electrical disturbance, resulting in a shift of the measured temperature. After shifting of the sensor's output by correcting its initial reading (initial temperature coincident with the other sensors), the numerical prediction agrees fairly well with the *in situ* measurement (maximum deviations of ~2 °C), so the cooling effect of the room temperature was satisfactorily captured.

The good agreement between the 2D and 3D FE model predictions leads to the conclusion that it is enough to use a 2D model for the present thermal analysis. Even so, the 3D model allows the analysis of the total concrete volume, providing a more complete representation of the temperature gradients (see Fig. 19). This advantage is particularly important, given the fact that the spiral case has a circular cross-section with a variable diameter. In another section, the 3D model detected a peak temperature value of 61 °C at the age of 1.8 days (see point P1 in Fig. 20).

According to the results obtained, the calculated temperatures are considered validated, so there is confidence to use them as inputs for the stress field computations.

3.5.2. Strains and stresses

Assessment of the mechanical model is herein performed by comparing the numerical predictions of the concrete total strains with the Carlson measurements, following a strategy analogous to the one adopted for the thermal field. However, here a huge difficulty is faced, as during early ages there is no reported robust methodology for zeroing the strains assessed with traditional strain gages sensors (Carlson ones included). This is due to two reasons: (i) the instant of total compatibility of the sensor with concrete is extremely difficult to be determined; and (ii) the Carlson sensor itself, made up of steel, induces a disturbance on the local strain to be measured. Thus, during the early ages the concrete strains provided by Carlson gages is unquestionably affected by

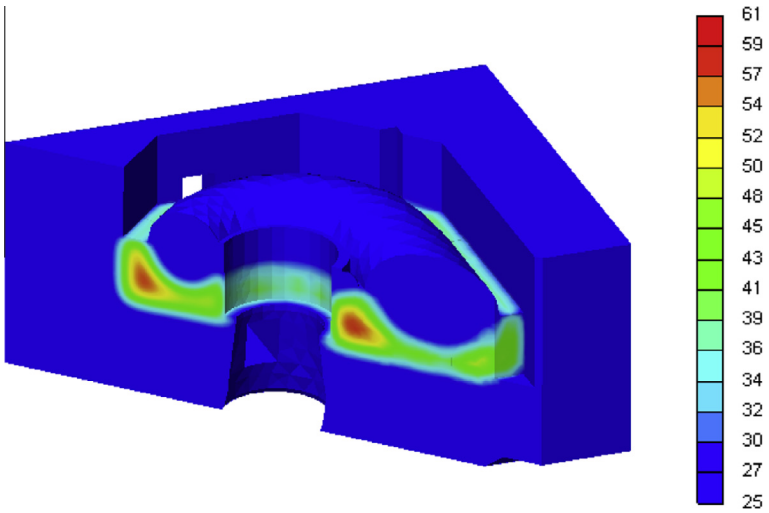


Fig. 19. 3D numerical temperature field at the age of 1.4 days [°C].

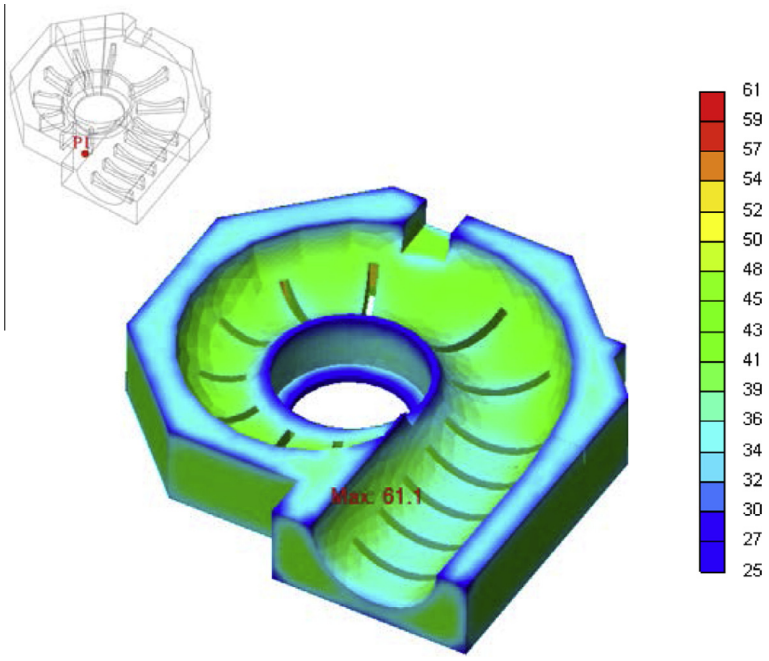


Fig. 20. 3D numerical temperature field at the age of 1.8 days [°C].

disturbances, which have to managed with some kind of engineering approach. The one adopted here for zeroing the Carlson strains consists in shifting vertically each sensor strain measurement curve, so that the corresponding peak could match the one predicted numerically. That peak approximately coincides with the instant at which concrete starts to cool down, and thus full compatibility of the sensor to the concrete is already accomplished (for further details on this strategy the reader is addressed to Ref. [48]). Comparison of the evolution trends of the numerical and monitored strains can then be performed straightforwardly, as at least the 'incremental strains' of concrete after the instant of 'full compatibility' of the strain gage to the concrete are well captured.

According to this methodology, the numerical and monitored strain evolutions are plotted together in Fig. 21 for the sensors S1, S2, S4, S5 and S6. Between the numerical and the experimental curves there is a good match for sensors S1, S4 and S5, and clearly poorer for the cases of sensors S2 and S6. As mentioned above, sensor S6 was potentially affected by an electrical disturbance, so its measurements may be misleading due to that unpredictable exogenous incident; they are therefore disregarded in further discussions. As far as sensor S2 is concerned, it can be observed that during the first 4 days the model predictions for the incremental concrete total strains are quite coherent with the measured ones, but from then onwards the numerical model underestimates the monitored strain drop tendency. This is probably related to a lower restriction to the deformation caused by the neighbouring concrete, or by a crack that might have occurred close to sensor S2.

The 3D model numerical predictions of the concrete principal stresses σ_1, σ_2 and σ_3 in the neighbourhood of sensors S2 and S6 are reproduced in Fig. 22, where it can be seen that concrete tensile strength f_{ct} evolution curve was never reached. Nevertheless, it should be pointed out that sensor S3 (in the 'no-stress box') was placed 0.10 m above sensor S2, and this particular detail was not introduced in the numerical model; so the real concrete strain

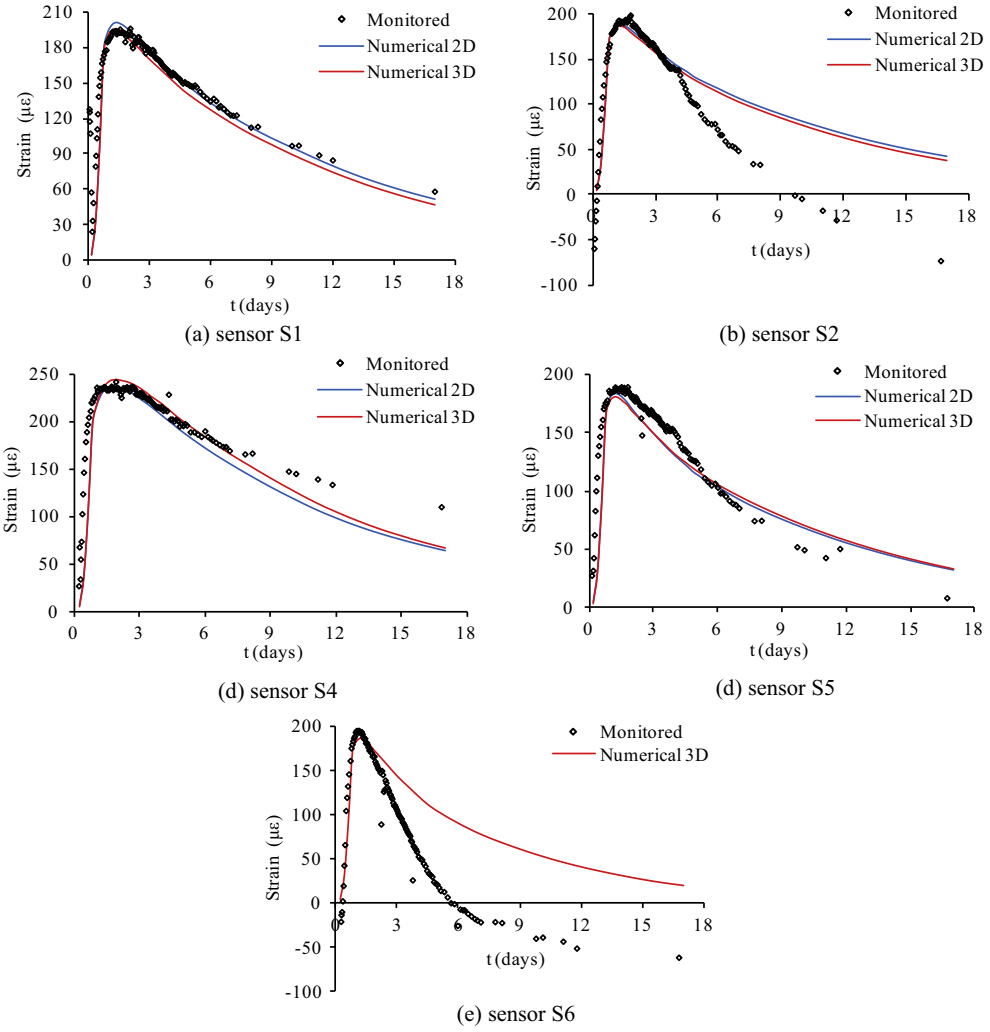


Fig. 21. Concrete total strains: numerical vs. monitored.

and stress states close to sensor S2 may have been affected by its proximity to the 'no-stress box'.

As previously noted, it is worth highlighting the difficulty of Carlson sensors to measure the concrete total strains during the early ages, since they depend on the full anchorage to this material. As sensors S2 and S6 were almost certainly affected by undesirable disturbances, comparison of the strain development tendencies predicted by the numerical model with the ones obtained from the monitoring campaign can only be properly made for sensors S1, S4 and S5. For these sensors the plots of Fig. 21 indicate that the evolution trends of the monitored concrete strain variations were satisfactorily captured by the numerical model. Besides being influenced by these strains, concrete stresses depend as well on the creep model (which was duly calibrated based on experimental testing) and on the thermal and autogenous deformations – the former depending on the local thermal information and on α_{cT}, and the latter having been obtained from lab characterization. Therefore, all the requirements for the adopted numerical model to provide satisfactory predictions of the concrete stresses are accomplished (at least as much as they can be, regarding the fact that an existing structure is being analysed, and several uncertainties are involved in the real scenario).

A good consistency between the axisymmetric and the 3D numerical results is found in Fig. 23, where evolutions of the normal stresses σ_z (circumferential) in both models are plotted together. Maps showing the temporal evolution of the σ_x (radial),

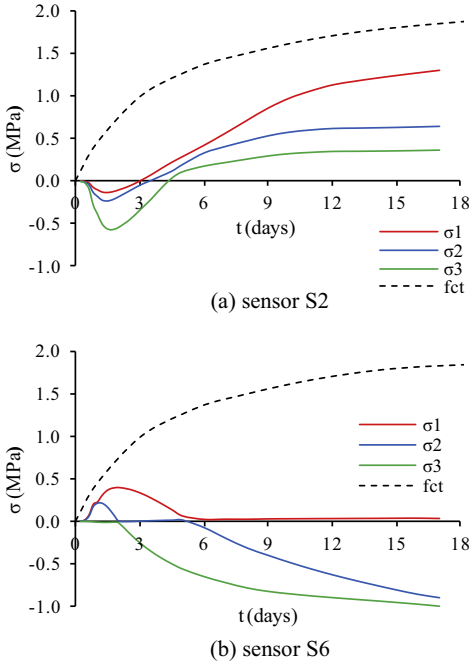


Fig. 22. Evolution of the concrete principal stresses and of the concrete tensile strength (3D model).

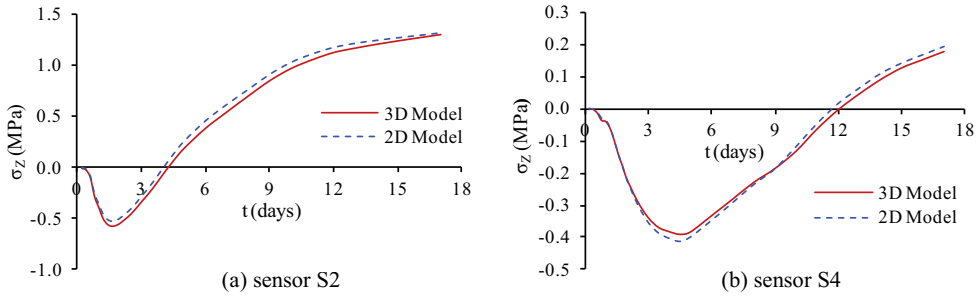


Fig. 23. Concrete normal stresses along direction *Z* (circumferential), predicted by the 2D and 3D models.

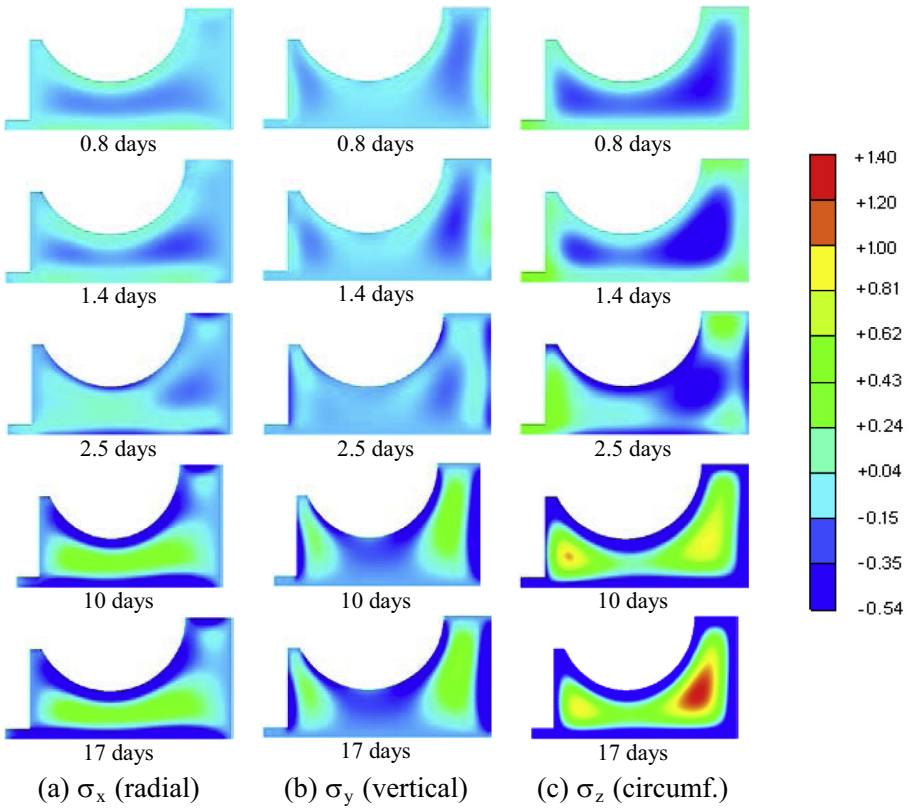


Fig. 24. Concrete normal stresses (MPa) in the 2D model.

σ_y (vertical) and σ_z (circumferential) stresses in concrete are presented in Fig. 24 for the 2D model. It can be seen that the concrete stresses vary significantly, due to the thermal field evolution along cement hydration. During the heating phase, which lasts until the age of 1.4 days, the volumetric expansion is higher in the core than in the surfaces, so the latter ones restrain the thermal expansion of the former. Thus, tensile stresses arise near the surfaces and compressive stresses appear in the concrete core. In the cooling phase, between the ages of 1.4 and 17 days, the volumetric contraction is larger in the core than that of the surfaces, to such an extent that the stress signs end up reversing, so that compressive stresses develop in the surfaces and tensile stresses occur in the core (it is worth remarking that concrete E-modulus is considerably higher at this stage). This behaviour was already reported in Fig. 22 for sensors S2 (core) and S6 (surface): close to sensor S2 the maximum concrete principal stress reaches 1.3 MPa at the age of 17 days, while at sensor S6 the peak tensile stress is 0.4 MPa at the age of 1.8 days. It should also be observed in Fig. 24 that normal stresses σ_z are much larger than those along *X* and *Y* directions, so the *Z* direction is the most unfavourable in terms of volumetric restraints.

Finally, evolution of the concrete cracking risk *CR* is depicted in Fig. 25 for two relevant instants. This figure also provides information on the percentage of concrete volume that is comprehended within each range of *CR* values (e.g., at the age of 0.8 days 35.5% of the concrete value has $0\% < CR < 10\%$). A $CR(t) = 1.0$ means that the maximum principal stress reached the concrete tensile strength, both at the same instant *t*. Emborg [49] suggests a maximum $CR = 0.7$ as suitable to ensure a reasonably low level of cracking on a concrete structure, taking into account the uncertainties associated to early age concrete behaviour, as well as to the unavoidable simplifications introduced in the mechanical analyses. On the other hand, the JCI Guidelines suggest a more restrictive maximum around $CR = 0.54$, which corresponds to a thermal cracking probability of 5% [30].

According to Fig. 25a it is possible to observe that *CR* is higher than 0.7 in only 0.7% of the concrete volume at the age of 0.8 days, and threshold $CR = 1$ is never reached. As far as the *CR* reported in

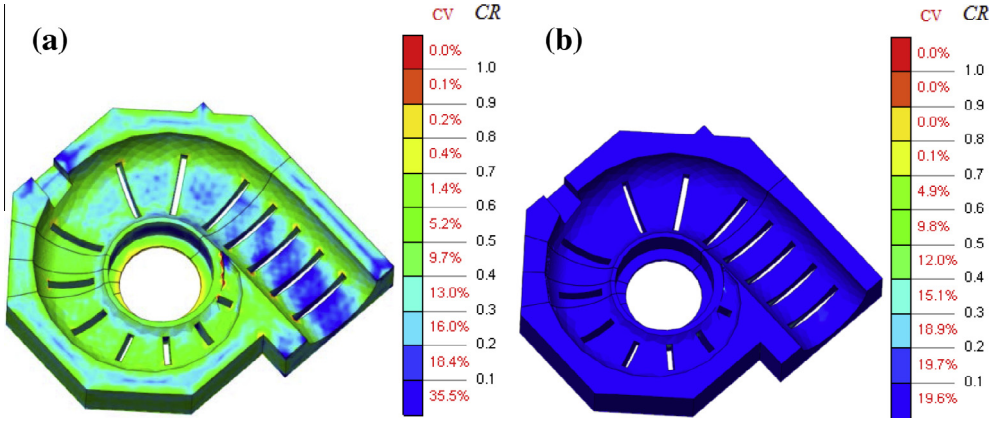


Fig. 25. Concrete cracking risk *CR*, at two ages, plotted as a function of its percentage of occurrence within the concrete volume (CV): (a) 0.8 days; (b) 17 days.

Fig. 25a, it must be emphasised that the peak values observable at this age are mostly induced by stress concentrations at several singular 'corner' points located at the 'skin' of the FE mesh (see e.g. Fig. 2b), which are unavoidable and induce spurious numerical stress peaks. This potential 'skin' cracking is negligible, since it tends to disappear subsequently, due to the compressive stress state that is generated in the surfaces upon cooling of the concrete core, but about 6.6% of the concrete volume is predicted to have a *CR* ranging from 0.5 to 0.7. Fig. 25b shows that at the age of 17days only 0.1% of the concrete volume presents $CR \geqslant 0.7$, but a considerable volume of concrete (~14.7%) is affected by a *CR* between 0.5 and 0.7. Finally, it is to be remarked that despite the considerably thermal cracking probability found in some regions of the concrete Layer 1, the structure did not exhibit any evidence of early age cracking.

4. Conclusion

The thermo-mechanical analysis of the concrete surrounding a turbine spiral case of the Batalha Hydroelectric Power Plant (Brazil) was presented in this paper, focused on the first casted concrete layer, and during a period of 17 days. This case study has allowed outlining the early age concrete behaviour of this particular type of structure. To do so, several laboratory tests were performed to characterize the concrete thermal and mechanical properties, and *in situ* measurements of temperatures and strains at the site were made using Carlson sensors.

The modelling methodology was briefly reported, with a special emphasis on the choice and definition of the material properties and boundary conditions. Based on the numerical results from 2D and 3D FE models, the following conclusions can be outlined:

- The adopted thermal model proved to be adequate for predicting the measured concrete temperatures, taking into account the cement heat of hydration release and the heat fluxes through the boundaries. The predicted temperatures agreed fairly well with the monitoring data, both in terms of the evolution tendencies and in matching the peak values. No significant differences between the 2D and the 3D models were observed.
- The explicit consideration of the water in the thermal model, involving an effective thermal conductivity greater than the one for stagnant water, and calibrated to account for the movement of water inside the spiral due to the thermal gradients, proved to be a satisfactory strategy for the numerical analyses.
- The monitored concrete strain variations were fairly well predicted by the numerical model, and the computed strains and stresses showed a good consistency in both the 2D and the 3D models. It is important to highlight the difficulty in defining the instant at which the strain gages are fully anchored to the hardening concrete, and start to measure concrete strains in a robust manner.
- The cracking risk *CR* was evaluated on the first layer of the concrete structure. Considerably high values of *CR* were numerically predicted in limited regions (partially related to stress concentrations, induced by singularities of the FE mesh), but the great majority of the concrete volume remained with a cracking risk below 0.7.

Acknowledgements

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References

[1] Noorzai J, Bayagoob K, Thanaoon W, Jaafar M. Thermal and stress analysis of Kinta RCC dam. Eng Struct 2006;28:1795–802.
[2] Liu J, Qiao L, Li P, Chun-Fang D. Laboratory test and numerical simulation of time-dependent thermomechanical behavior of the three-gorges dam. J Mater Civ Eng 2010;22(2):111–23.
[3] Kuzmanovic V, Savic L, Mladenovic N. Computation of thermal-stresses and contraction joint distance of RCC dams. J Therm Stresses 2013;36:112–34.
[4] ICOLD. Thermal analysis of a RCC dam body during construction. In: 7th Benchmark workshop on numerical analysis of dams. Bucharest, Romania; 2004.
[5] ICOLD. Simulation of the thermo-mechanical effects occurring during the construction phases of a concrete arch dam. In: 10th Benchmark workshop on numerical analysis of dams. Paris, France; 2009.
[6] Fairbairn E, Silvoso M, Koenders E, Ribeiro F, Filho R. Thermo-chemo-mechanical cracking assessment for early-age mass concrete structures. Concr Int 2012;34(5):30–5.
[7] Benboudjema F, Torrenti J. Early-age behaviour of concrete nuclear containments. Nucl Eng Des 2008;238:2495–506.
[8] Concrack 3: Control of Cracking in Concrete Strutures 3. JCI-RILEM international workshop on crack control of mass concrete and related ussues concerning early-age of concret strutures. Paris, France; 15–16 March, 2012.
[9] EM 1110-2-3001. Engineering and design – planning and design of hydroelectric power plant structures. Washington, DC: U. S. Army Corps of Engineers; 1995. 36p.
[10] Sharda K, Lakshminarayan P, De J, Krishna K. Thermal analysis of draft tube and spiral mass concrete. Indian Concr J 2004;78:11–6.
[11] Xu X, Ma Z, Zhang H. Research on transient temperature of concrete surrounding spiral case and temperature effect on gaps in a hydropower station. Adv Mater Res 2011;163–167:1724–7.
[12] Xu X, Luo Q, Ma Z. Numerical simulation of temperature and creep of concrete surrounding spiral case in a hydraulic plant. Adv Mater Res 2011;194–196:977–80.

[13] Jun Z, Ya-hui D, Jia-ming W. Temperature control research on spiral case concrete of Xiluodu underground power plant during construction. Appl Mech Mater 2013;328:b933–41.
[14] Faria R, Azenha M, Figueiras JA. Modelling of concrete at early ages: application to an externally restrained slab. Cement Concr Compos 2006;28(6):572–85.
[15] Azenha M, Faria R. Temperatures and stresses due to cement hydration on R/C foundation of a wind tower – a case study. Eng Struct 2008;30:2392–400.
[16] Azenha M, Lameiras R, Sousa C, Barros J. Application of air cooled pipes for reduction of early age cracking risk in a massive RC wall. Eng Struct 2008;30:2392–400.
[17] Çengel Y, Turner R. Fundamentals of thermal-fluid sciences. 2nd ed. McGraw-Hill; 2005.
[18] Reinhardt HW, Blaauwendraad J, Jongedijk J. Temperature development in concrete structures taking account of state dependent properties. In: Int conf concrete at early ages, Paris, France; 1982.
[19] Breugel K. Simulation of hydration and formation of structure in hardening cement-based materials. Delft University of Technology. PhD Thesis; 1991.
[20] Bentz D. Three-dimensional computer simulation of Portland cement hydration and microstructure development. J Am Ceram Soc 1997;80(1):3–21.
[21] Maekawa K, Chaube R, Kishi T. Modelling of concrete performance: hydration, microstructure formation, and mass transport. E & FN Spon; 1999.
[22] Wadsö L. An experimental comparison between isothermal calorimetry, semi-adiabatic calorimetry and solution calorimetry for the study of cement hydration. Nordtest report TR 522; 2003.
[23] Azenha M. Numerical simulation of structural behavior of concrete since its early ages. Faculdade de Engenharia da Universidade do Porto. PhD Thesis; 2009.
[24] Incropera FP, DeWitt DP. Introduction to heat transfer. New York: John Wiley & Sons; 2001.
[25] Carino N, Lew N. The maturity method: from theory to application. Washington, DC: Structures Congress & Exposition; 2001.
[26] Bazant Z, Osmand E. Double power law for basic creep of concrete. Mater Struct, Res Test 1976;9:3–11.
[27] TNO Diana BV 2011, DIANA User's Manual – Release 9.4.4.
[28] Choi S, Won M. Thermal strain and drying shrinkage of concrete structures in the field. ACI Mater J 2010;107(5):498–507.
[29] Yeon J, Choi S, Won M. In situ measurement of coefficient of thermal expansion in hardening concrete and its effect on thermal stress development. Constr Build Mater 2013;38:306–15.
[30] JCI, Japan Concrete Institute. JCI guidelines for control of cracking of mass concrete; 2012.
[31] Lee Y, Choi M, Yi S, Kim J. Experimental study on the convective heat transfer coefficient of early-age concrete. Cement Concr Compos 2009;31:60–71.
[32] AssociaçãoBrasileira de Normas Técnicas. NBR 5735: Blast fumace slag Portland cement – Specification. Rio de Janeiro, Bazil; 1991 [in Portuguese].
[33] Wang X, Lee H. Simulation of a temperature rise in concrete incorporating fly ash or slag. Mater Struct 2010;43:737–54.
[34] Schutter G. Hydration and temperature development of concrete made with blast-furnace slag cement. Cem Concr Res 1999;29:143–9.
[35] Breugel K. Artificial cooling of hardening concrete. Concrete structures. Delft: Delft University of Technology; 1980.
[36] Associação Brasileira de Normas Técnicas. NBR 12818: Concrete – Determination of thermal diffusivity – Test method. Rio de Janeiro, Brazil; 2012 [in Portuguese].
[37] Associação Brasileira de Normas Técnicas. NBR 8222: Concrete – Determination of the elasticity modulus by compression. Rio de Janeiro, Brazil; 2008 [in Portuguese].
[38] Associação Brasileira de Normas Técnicas. NBR 7222: Concrete and mortar – determination of the tension strength by diametrical compression of cylindrical test specimens. Rio de Janeiro, Brazil; 2011 [in Portuguese].
[39] Associação Brasileira de Normas Técnicas. NBR 8224: Hardened concrete – determination of creep – test method. Rio de Janeiro, Brazil; 2012 [in Portuguese].
[40] Baroghel-Bouny V, Mounanga P, Khelidj A, Loukili A, Rafai N. Autogenous deformations of cement pastes Part II. W/C effects, micro-macro correlations, and threshold values. Cem Concr Res 2006;36:123–36.
[41] Bjøntegaard Ø, Sellevold E. Autogenous and thermal deformations. Report 25, A. Bentur, RILEM publications s.a.r.l; 2002. p. 65–88.
[42] Laplante P, Boulay C. Evolution du coefficient the dilatation thermique de béton en fonction de sa maturité aux tout premier âges. Mater Struct 1994;27:596–605.
[43] Cusson D, Hoogeveen T. Measuring early-age coefficient of thermal expansion in high-performance concrete. In: Int RILEM conf on volume changes of hardening concrete: testing and mitigation, Lyngby, Denmark; 2006.
[44] Viviani M, Glisic B, Smith I. Separation of thermal and autogenous deformation at varying temperatures using optical fiber sensors. Cement Concr Compos 2007;29:435–77.
[45] Mottaghy D, Hans-Dieter V, Schellchmidt R. Temperature dependence of the relationship of thermal diffusivity versus thermal conductivity for crystalline rocks. Int J Earth Sci 2008;97(2):435–42.
[46] Malafaia A. Geomechanics classification systems applied to rock slopes of the UHE Batalha. Universidade Federal do Rio de Janeiro. MSc Thesis; 2007 [in portuguese].
[47] Panigrahi R, Dhar Y. Geotechnical evaluation for the stability of underground powerhouse cavity at Theri Dam project, Garhwal Himalayas, India. In: ISRM Symposium, Eurock '92, Chester, UK; 1992.
[48] Azenha M, Faria R, Ferreira D. Identification of early-age concrete temperatures and strains: monitoring and numerical simulation. Cement Concr Compos 2009;31:369–78.
[49] Emborg M. Models and methods for computation of thermal stresses. Report 15, R. Springenschmid, RILEM publications s.a.r.l; 1998. p. 178–230.